

Chapter 9 continued

Questions

1 E.g.

Data type	Example
Boolean	True
Integer	1
Text	"Hello"
Character	"!"
Real	2.36
Date/time	5/5/2021

2 A table has several fields. Each field is one piece of information. A row of data about one object/item is called a record.

3 To uniquely identify each record

4 a E.g.

Field name	Data type
Username	Text/alphanumeric
Password	Text/alphanumeric
Online name	Text/alphanumeric
Date account created	Date/time
Number of friends	Integer

b E.g. Username, or adding an ID, e.g. user ID

5 12GH

59RF

3WR

52D

6SE

6 Saturn Chocolate

Vanilla Fudge

Peanuts Nuts

Jazz Fruit

Signet Fudge

7 SELECT **Cost**

FROM PRODUCTS

8 **SELECT** ID Number, **Name**

FROM PRODUCTS

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9 **SELECT** Name, Cost, Quantity in stock
FROM PRODUCTS

10 596 Computer Science Programming 2
806 Computer Science Programming 2

11 596 Computer Science Pass
123 Maths Pass
023 Maths Pass
806 Computer Science Pass
596 Science Pass

12 **SELECT** Test name, Percentage, Grade
FROM MARKS
WHERE Percentage > 50

13 **SELECT** Student ID, Subject, Percentage
FROM MARKS
WHERE SUBJECT = "Science"

14 **SELECT** Student ID, Test name, Percentage
FROM MARKS
WHERE Percentage <= 25

15

20.3	22	11/12/2019
23.7	15	11/12/2019

16

11/12/2019	13:00	63
11/12/2019	15:00	61
11/12/2019	18:00	60

17

11/12/2019	10:00
11/12/2019	13:00
11/12/2019	15:00
11/12/2019	18:00
12/12/2019	15:00

18 **SELECT** Time
FROM WEATHER
WHERE Date = 12/12/2019 **AND** Humidity > 50

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19 **SELECT** Wind speed, **Temperature**

FROM WEATHER

WHERE Wind speed < 15 **OR** Wind speed > 20

20 17.2 18 12/12/2019

18.4 20 12/12/2019

20.3 22 11/12/2019

23.7 15 11/12/2019

23.9 16 11/12/2019

24.0 13 11/12/2019

21 11/12/2019 13:00 23.7 63

11/12/2019 15:00 24.0 61

11/12/2019 18:00 23.9 60

11/12/2019 10:00 20.3 59

12/12/2019 15:00 18.4 52

12/12/2019 18:00 17.2 48

22 **SELECT** Date, Wind speed, Humidity

FROM WEATHER

ORDER BY Wind speed

23 **SELECT** Date, Time, **Temperature**

FROM WEATHER

ORDER BY **Humidity DESC**

24 **SELECT** **Temperature**, Wind speed, **Humidity**

FROM WEATHER

ORDER BY **Temperature DESC**

25 51

26 8

27 1

28 135.71

29 **SELECT** **COUNT**(FirstName)

FROM ORDERS

WHERE FirstName = "James" **AND** LastName = "Smith"

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```
30 SELECT SUM(TotalCost)
    FROM ORDERS
    WHERE NumberItems >= 5
```

```
31 SELECT COUNT(NumberItems)
    FROM ORDERS
    WHERE TotalCost > 50.0
```

Skills focus 9.1 questions

- 1 Sum adds together the values. Count identifies how many records there are.
- 2 160
- 3 5
- 4 2.78
- 5 4

Exam-style questions

- 1 One mark per definition.

Record the completed fields for one object / item. Example: all the data about one student.

Field one piece of data. Example: Date of birth.

- 2 One mark per description and example to max four.

Boolean: can only be 2 values Example: true.

Real: decimal number. Example: 23.5.

Date/time: a date or a time. Example: 1/1/1900.

Text: one or more letters, symbols and/or numbers. Example: "hello!".

- 3 One mark per bullet.

- ID (or other appropriate key field), data type Integer / Text
- Screen number, data type Integer / Text
- Film, data type Text / Alphanumeric
- Date, data type Date / Time
- Time, data type Date / Time
- SeatsSold, data type Integer
- Primary key is ID (or other appropriate key field defined).

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4 a 6

b 7

c One mark for definition, e.g. A unique identifier.

One mark for Appointment ID.

d One mark for first correct, one mark for remainder correct.

JP	Cut
AD	Cut and dry
AD	Colour
JP	Cut and dry
JP	Cut
AD	Colour

e 2

f One mark for each statement.

```
SELECT Customer first name, Customer last name
FROM APPOINTMENTS
WHERE Date = 1/2/2020
```

g One mark for each statement.

```
SELECT COUNT(Employee ID)
FROM APPOINTMENTS
WHERE Employee ID = "JP"
```

5 a One mark per bullet.

- Sales ID: integer
- Date: date / time
- Customer ID: text
- Quantity bought: integer
- Total Cost: real

b One mark for one correct, two marks for all correct in the correct order.

3/3/2020	1GHF2
3/3/2020	1GHF2
4/4/2020	1GHF2

c One mark for each completed statement.

```
SELECT Customer ID, Total Cost
FROM SALES
WHERE Total Cost > 1
```

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d One mark for **one** correct, **one** mark for all correct and in the correct order.

12 151
13 28

e One mark for each completed statement.

```
SELECT SUM(Quantity bought)
FROM SALES
WHERE Date = 4/4/2020
```

f One mark for each completed statement.

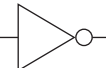
```
SELECT SUM(Total Cost)
FROM SALES
```

Chapter 10: Boolean logic

Discussion questions

- 1 In Boolean, values are either true (1) or false (0). In a computer, if there is electricity that means 1 and if there is no flow of electricity that means 0.
- 2 A logic gate takes one or more inputs as a Boolean value (or values) and outputs a value.
- 3 All digital computers represent data in binary form with 0s and 1s, so they would not be able to operate without logic gates.

Questions

- 1 Input  Output

2

A	X
1	0
0	1

- 3 $X = \text{NOT } A$

- 4 Input 1  Output
Input 2

5

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

- 6 $X = A \text{ AND } B$